



Essential new PLS-SEM analysis methods for your entrepreneurship analytical toolbox

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Abstract

Rigorous research across the wide spectrum of entrepreneurship is important for developing knowledge and innovation in the field, which in turn offers building blocks for future inquiry, practice, innovation, and economic growth. For this reason, scholars continually seek new empirical analysis tools to examine constructs and variables included in their research. While many familiar tools remain at the core of the scholar's tool box, PLS-SEM offers valuable options that have been accentuated by recent developments. This article provides a thorough review of PLS-SEM and its application to current entrepreneurship research, summarizes the emergent confirmatory composite analysis (CCA) processes for applying and interpreting the analysis, explains why PLS-SEM can be especially advantageous for research in this field, and concludes with suggestions for scholars as they plan future research projects. In addition, recent PLS-SEM developments are shared that may effectively apply to entrepreneurship research.

Keywords PLS-SEM · Entrepreneurship research · Empirical analysis · Quantitative tools · Confirmatory composite analysis

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Introduction

For many years, entrepreneurship scholars have relied on multiple regression as their data analysis tool of choice. Indeed, we estimate that 75% of empirical studies in entrepreneurship apply multiple regression to analyze the relationships between their study variables.¹ Multiple regression, along with other basic techniques like correlation and Chi-square, has served entrepreneurship researchers well in understanding and reporting their findings. To enhance the body of entrepreneurship knowledge and to better equip researchers for an academic landscape in which publishing requirements are quickly evolving, we provide an overview of several new tools scholars need to add to their analytical toolbox.

Among the more promising analytical tools that have emerged in the past decade is partial least squares structural equation modeling (PLS-SEM). Scholars who earned their Ph.D. in the last 30 years likely learned the basics of covariance-based structural equation modeling (CB-SEM). For too many of them, however, their experience with CB-SEM was likely frustrating due to the challenges of obtaining an acceptable solution. That is, do the measurement and structural models meet all of the recommended guidelines for goodness of fit (GoF)? Rather than GoF, the primary metric in assessing models using PLS-SEM is the prediction of explained variance in the dependent variable(s), a similar metric to multiple regression. Thus, if you understand the basics of multiple regression, then you too can apply PLS-SEM.

The purpose of this article is to provide an overview of PLS-SEM for entrepreneurship scholars. We first report the results of our search of entrepreneurship journals aimed at identifying the frequency of PLS-SEM applications. Then we define and explain the major aspects of PLS-SEM and the metrics used to evaluate the results. Next, we summarize a step-by-step process for applying the method and interpreting the findings. We then suggest potential applications of PLS-SEM in entrepreneurship research, and we conclude by summarizing the advantages and limitations of this new analytical tool. In addition, we share recent PLS-SEM developments applicable to entrepreneurship research.

Application of PLS-SEM in entrepreneurship research

To assess the use of PLS-SEM in entrepreneurship research, we used Google Scholar to search peer-reviewed entrepreneurship journals. In executing this search, we included published articles with “PLS-SEM” in the text, and constrained our search to journals whose titles included the following terms: venturing, entrepreneurship, entrepreneur, and entrepreneurial (see appendix). Our search did not produce any entrepreneurship studies applying PLS-SEM prior to 2011. Consequently, our results reflect 2011 through July of 2019. This process identified the number of studies using PLS-SEM in each of the entrepreneurship journals we reviewed. Although we did not perform an

¹ Using ABI/Inform, we conducted a search of peer reviewed articles including “entrepreneur” or “entrepreneurship” in the article title, and that search produced 10,228 pieces. We next limited that search to articles including “multiple regression” anywhere in the article, and that constriction reduced the findings to 7802 pieces, or 76.28% of the original search results. This supports our estimate that approximately 75% of entrepreneurship studies in peer reviewed journals apply multiple regression.

in-depth review of every article, we confirmed that each study found in our search did apply PLS-SEM. Table 1 summarizes the number of peer-reviewed articles found in our search per year for each keyword applied. Figure 1 graphically demonstrates the total number of articles applying PLS-SEM in entrepreneurship journals each year and the cumulative total.

A wide variety of entrepreneurship journals have published studies applying PLS-SEM. Although the number of studies is quite small, it is growing rapidly. As with other statistical tools, the increased utilization of PLS-SEM is likely due to the emergence of user-friendly graphical interface software, such as SmartPLS (Ringle et al. 2015) and PLS Graph (Chin 2003), as well as relatively understandable textbooks such as Hair Jr et al. (2016) and Hair Jr et al. (2019a). Our analysis reveals most studies applying PLS-SEM in entrepreneurship were published in the last four years (95% of the total), so we consider this method to be at an embryonic stage in entrepreneurship research.

Several peer-reviewed journals in other disciplines have published articles reviewing PLS-SEM applications in their fields. Examples include information systems (Hair Jr., Hollingsworth et al. 2017), hospitality (Ali et al. 2018), marketing (Hair Jr et al. 2011; Hair Jr et al. 2012a), management accounting (Nitzl 2016), strategic management (Hair Jr et al. 2012b), international business (Richter et al. 2016b), human resource management (Ringle et al. 2018), and family business (Sarstedt et al. 2014b). However, no studies exist examining the application of PLS-SEM in entrepreneurship research. Given that gap, the growing application of PLS-SEM, and the method's logical applicability for entrepreneurship research, we next present an overview of PLS-SEM and then discuss its “silver bullet” (Hair Jr et al. 2011, p.139) potential for entrepreneurship research.

What is PLS-SEM: An overview

The choice of structural equation modeling (SEM) as an analytical method is based on the need to test complex theories and concepts (Rigdon 1998). Much of the popularity of

Table 1 Number of peer-reviewed articles found in our search per year for each journal title keyword applied

Year	Key word				Total
	Venturing	Entrepreneur	Entrepreneurship	Entrepreneurial	
2011			2		2
2012			1		1
2013	1		2		3
2014			5		5
2015			12		12
2016			21	1	22
2017	2		21	2	25
2018	2	1	44	2	49
2019 through July	1		47	6	54
Totals	6	1	155	11	173

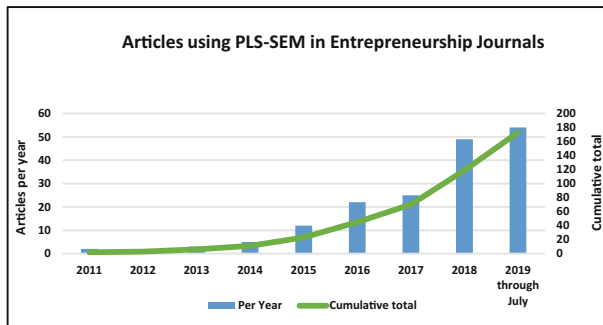


Fig. 1 Number of articles in entrepreneurship journals applying PLS-SEM

SEM can be attributed to the method's ability to simultaneously execute two steps: (1) evaluate the measurement quality of multi-item latent variables, and (2) test the relationships between multi-item latent variables (Babin et al. 2008). While initial applications of structural equation modeling utilized a covariance-based approach (CB-SEM), which required two rather complex steps. Researchers now have the option of applying the single-step variance-based PLS-SEM. Simply stated, PLS-SEM simultaneously examines the relationships between multi-item latent variables. The statistical objective is to maximize the explained variance in the one or more multi-item dependent variables. In PLS-SEM, the latent variables scores, which are calculated based on total variance, are referred to as composite variables. With CB-SEM, the latent variables are calculated based on common variance only and are referred to as common factor variables.

Much more than regression

Structural equation modeling, and particularly PLS-SEM, is an extension of multiple regression. While multiple regression is limited to a single dependent variable and several single-indicator independent variables, PLS-SEM can easily handle two or more multi-indicator dependent variables and many multi-indicator independent variables. Figure 2 presents a PLS-SEM model recently applied in entrepreneurship research assessing agriculture students' cultural values and their

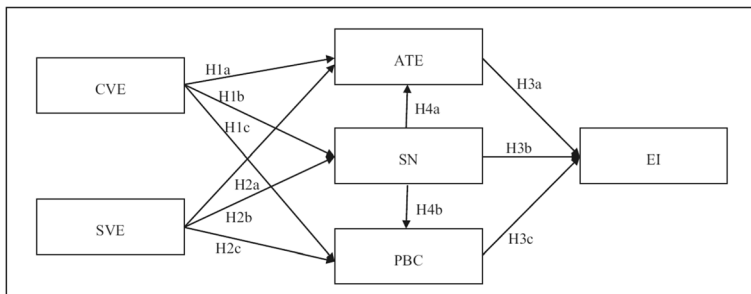


Fig. 2 An example of a somewhat complex PLS-SEM model from Shiri et al. (2017) study of agriculture students' cultural values and their entrepreneurial intentions

entrepreneurial intentions (Shiri et al. 2017). The model includes two independent variables, social valuation of entrepreneurship (SVE) and closer valuations of the environment (CVE). Each of those two independent variables are hypothesized as having a positive effect on three dependent variables: personal attitude (ATE), subjective norm (SN), and perceived behavioral control (PBC). ATE, SN, and PBC are hypothesized as having a positive relationship with entrepreneurial intention (EI). Further, SN is hypothesized as having a positive relationship with ATE and PBC. As is common in both PLS-SEM and entrepreneurship research, multiple variables were measured with more than one indicator (survey item). To fully understand this model and its theoretical foundation, the reader should examine Shiri et al. (2017). Thus we argue that PLS-SEM is a powerful tool through which entrepreneurship researchers can simultaneously assess the relationships between multiple variables that are measured with more than one item, an approach not possible with multiple regression.

When and why use PLS-SEM

PLS-SEM was originally developed by Herman Wold (1982), who initially referred to the method as PLS path modeling. More recently scholars branded the method as PLS-SEM (Hair Jr et al. 2011). Methodologically speaking, PLS-SEM solutions are obtained by estimating partial components of structural models and combining principal components analysis with ordinary least squares regressions (Mateos-Aparicio 2011). Because CB-SEM (Jöreskog 1973) has numerous restrictive assumptions (Hair Jr et al. 2011), PLS-SEM is typically viewed as an effective alternative. Stated again, PLS-SEM – sometimes referred to as variance-based SEM –has fewer restrictive assumptions than CB-SEM (Hair Jr et al. 2017a).

From the points above and the materials that follow, we provide the following guidelines to assist entrepreneurship researchers in determining whether PLS-SEM is the appropriate method to apply. The method is appropriate in any of the following contexts, when:

- 1) prediction is the primary statistical objective of the research;
- 2) conducting exploratory research to develop or extend theory;
- 3) the research includes multi-item latent variables;
- 4) one or more of the latent variable measurement models are measured formatively;
- 5) the scaling method is non-parametric, such as ordinal or nominal measurements;
- 6) the data consists of financial ratios or similar types of secondary/archival data;
- 7) the research focus is on a small population sample that results in a small sample size, such as in business-to-business research;
- 8) large samples are available (10,000+);
- 9) the structural model is complex and includes many constructs, indicators, and/or causal relationships;
- 10) data are not normally distributed, as is typical of social sciences and survey data.

Faced with any of the above conditions, PLS-SEM is likely an appropriate method (Hair Jr et al. 2016).

Application of PLS-SEM: A two step process

Application of PLS-SEM first requires measurement model evaluation using confirmatory composite analysis (Hair Jr. et al. 2020). When all CCA rules of thumb are met or exceeded, the researcher then moves on the structural model assessment. We first discuss CCA and then evaluation of the structural model.

Measurement model evaluation (CCA) The first step in applying PLS-SEM is to evaluate the reliability and validity of the measurement model. In CB-SEM, this step is referred to a confirmatory factor analysis (CFA). This similar step in PLS-SEM is identified as confirmatory composite analysis (CCA) (Hair Jr et al. 2020; Hair Jr. et al. 2017a; Hair Jr. et al. 2019b; Hair Jr and Sarstedt 2019). The primary reason for referring to this step as CCA is because PLS-SEM is based on evaluating composite latent variables that use total variance (Hair Jr et al. 2020; Hair Jr et al. 2017a; Hair Jr et al. 2019b; Hair Jr and Sarstedt, 2019), which contrasts with the common CB-SEM factors using only common variance. Thus, CCA differentiates the PLS-SEM process of evaluating measurement models from the more familiar confirmatory factor analysis (CFA) approach utilized with CB-SEM.

CCA is a systematic methodological process to confirm measurement models in PLS-SEM. The steps differ based upon whether the measurement model is reflective or formative. For reflective measurement models, the steps are as follow: (1) estimate item loadings and assess significance, (2) evaluate indicator reliability, (3) assess composite reliability of the constructs, (4) examine average variance extracted (AVE), (5) confirm discriminant validity using the HTMT method, (6) assess nomological validity, and (7) evaluate predictive validity. The CCA steps for formative measurement models are somewhat different and include the following: (1) execute and assess convergent validity, (2) evaluate indicator multicollinearity, (3) evaluate the size and significance of indicator weights, (4) examine the relative contribution of the indicators by reviewing the size and significance of their loadings, and (5) confirm predictive validity.

For an early application of the CCA process, refer to Hair Jr., Manley, Williams, and Forrester (2019). Table 2 reiterates these basic CCA steps and rules of thumb for both reflective and formative models. For a more detailed description of the CCA steps, see Hair Jr et al. (2020). If the recommended rules of thumb are met for each of the steps – whether the model is formative or reflective – the next step is to evaluate the structural model.

Structural model assessment A systematic process is also followed to evaluate a structural model in PLS-SEM with the following steps: (1) evaluate multicollinearity between the independent variable constructs of the structural model; (2) examine the size and statistical significance of the path coefficients; assess in-sample prediction of the dependent constructs based on (3) the R^2 of the endogenous variable(s), (4) the effect size (f^2), and (5) the predictive relevance (Q^2); and (6) evaluate the out-of-sample predictive validity using PLSpredict (Shmueli et al. 2019b). If the recommended rules of thumb are met for measurement and structural models, then the researcher is ready to report the findings. Table 3 summarizes these basic structural model assessment steps and rules of thumb. For a more detailed description of the structural model assessment steps, see Hair Jr et al. (2020).

Table 2 Measurement model evaluation (CCA) steps and rules of thumb for reflective and formative modelsMeasurement Model Evaluation (CCA) – *Reflective*

<i>Steps</i>	<i>Rules of Thumb</i>
(1) Estimate item loadings and assess significance	Loadings >0.708 , and t-statistic $> \pm 1.96$
(2) Evaluate indicator reliability	Squared indicator loadings
(3) Assess composite reliability of the constructs	Cronbach's alpha, composite reliability, or $\rho_A > 0.70$
(4) Examine average variance extracted (AVE)	$AVE \geq 0.50$
(5) Confirm discriminant validity using the HTMT method	HTMT confidence intervals >0.85 (or 0.90)
(6) Assess nomological validity	Correlate the construct score of each construct with one or more other constructs in the nomological net (but not in the model being tested).
(7) Evaluate predictive validity	R^2 , f^2 (effect size), Q^2 (blindfolding)

Measurement Model Evaluation (CCA) – *Formative*

(1) Execute and assess convergent validity	Path coefficient to alternate (reflectively-measured) construct ≥ 0.70
(2) Evaluate indicator multicollinearity	VIF values ≤ 3.0
(3) Evaluate the size and significance of indicator weights	Assess size of weights to determine their individual importance, statistical significance ≤ 0.05
(4) Examine the relative contribution of the indicators by reviewing the size and significance of their loadings	Formative indicators must not be removed solely on statistical criteria
(5) Confirm predictive validity	R^2 , f^2 (effect size), Q^2 (blindfolding)

Table 3 Structural model assessment steps and rules of thumb

Structural Model Assessment

<i>Steps</i>	<i>Rules of Thumb</i>
(1) Evaluate multicollinearity between the independent variable constructs of the structural model	VIF values <3.0
(2) examine the size and statistical significance of the path coefficients	Higher path coefficients are more powerful predictors, statistical significance ≤ 0.05
(3) assess in-sample prediction of the dependent constructs based on the R^2 of the endogenous variable(s)	Higher R^2 values are more desirable
(4) assess in-sample prediction of the dependent constructs based on the effect size (f^2)	$0.02 < f^2$ value <0.15 – small; $0.15 < f^2$ value <0.35 – medium; and f^2 value >0.35 – large effect size
(5) assess in-sample prediction of the dependent constructs based on the predictive relevance (Q^2)	Q^2 values >0.0 – meaningful; $0.25 < Q^2$ values <0.50 – medium; Q^2 value >0.5 – large predictive relevance
(6) evaluate the out-of-sample predictive validity using PLSpredict	See Hair Jr.et al. (2020)

Applying PLS-SEM in entrepreneurship research Proper application of PLS-SEM will enhance and extend entrepreneurship research. We thus emphasize seven facets of PLS-SEM that are particularly relevant and applicable for entrepreneurship: (1) prediction of multiple endogenous constructs, (2) applicability with small and/or non-normally distributed samples, (3) ability to assess complex models, (4) capacity to apply both formative and reflective measures, (5) examination of mediation and conditional process models, (6) assessment of out-of-sample prediction using holdout samples, and (7) multi-group analysis and hierarchical component models. This section summarizes several advanced analysis algorithms available in PLS-SEM to test entrepreneurship theories and explore additional constructs.

Exploratory research predicting endogenous constructs

PLS-SEM is particularly effective in exploratory studies focused on prediction, which includes determining drivers of endogenous constructs (Hair Jr et al. 2011; Hair Jr et al. 2014). A valuable business research outcome is the knowledge gained that could enhance firm financial performance or other desirable practitioner outcomes (McDowell et al. 2019; Sarstedt et al. 2014a). Unfortunately, applicable and practical knowledge is often neglected in business research. Illustrating that the prediction of performance drivers is particularly important in practitioner research (Hair Jr et al. 2017c; Hair Jr. et al. 2019c), Merchant (2012) applies the phrase “[i]n settings that are like yours..., you can get the best results by doing this, and here is why it works” (p. 341).

Hindle et al. (2004) noted that entrepreneurship researchers have not placed appropriate emphasis on making scholarly findings relevant to practitioners. Although some time has passed since Hindle et al.’ (2004) assessment, it was accurate then and remains so now. Given that entrepreneurs need help in knowing how to “make things happen” (p.264), this void in practitioner-relevant entrepreneurship research is ironic. Such “things” might include not only improving firm performance, but also issues such as how best to motivate employees or which marketing practices are most effective (Watad et al. 2007) in entrepreneurial endeavors. As practitioner knowledge is best assimilated from a “holistic” view of the business environment (Merchant 2012, p. 345), PLS-SEM’s ability to facilitate exploratory, predictive, and complex models (discussed below) further enhances researchers’ ability to build applicable practitioner insights. Developing a comprehensive understanding of these types of topics involves applying a more complex method of statistical analysis, yet one with results easily communicated to practitioners. We believe the best available method is PLS-SEM.

The need for practitioner knowledge is not limited to just entrepreneurs. People working in government-sponsored entrepreneurship programs, support centers such as the small business development centers or women’s business centers, centers for closely held or family businesses, university entrepreneurship centers, private-sector consultants and advisors, and entrepreneurship educators all need relevant knowledge they can easily pass to diverse stakeholders (Hindle et al. 2004; Watad et al. 2007). Given PLS-SEM is useful in exploratory studies aimed at prediction, its application in entrepreneurship research may help generate useful practitioner knowledge for multiple audiences.

Exploratory research is particularly important when theory is less developed (Hair Jr et al. 2016). Theory development in the entrepreneurship field is still in its adolescence, often integrating theoretical concepts from other fields such as “sociology, psychology, anthropology, marketing, management, finance, organizational behavior, and engineering (Kuratko et al. 2015, p.1)”. In more developed fields, theory testing involves applying hypotheses formed from a “nomological net” of existing knowledge (Sarstedt et al. 2014a, p. 155). However, as researchers are still knitting entrepreneurship theory’s nomological web, exploratory research is a major component of entrepreneurship studies (e.g., Bohnsack et al. 2019; Daud et al. 2019; Liu 2019; Pandit et al. 2018; Srivastava et al. 2017; Patel 2019).

Exploratory research focuses on “what, when, why, and how” (Hair Jr. et al. 2017b, p. 447). In building entrepreneurship theory, researchers can apply PLS-SEM to address questions such as these: Why does someone pursue entrepreneurial endeavors? What environmental aspects, internal or external, prompt entrepreneurial success? At what stage in the entrepreneurial process should one expect to successfully solicit outside funding? How is entrepreneurial ambition developed in individuals? (These are but a few of the many possibilities.) In exploring causal relationships, such as those related to “what, when, why, and how” questions, researchers may provide associations that lead to entrepreneurship theory formation. However, as entrepreneurship researchers seek associations leading to theory development and explore newly perceived causal relationships, they need flexibility in modeling – the ability to easily move constructs and retest relationships through explained variance – and PLS-SEM accommodates that prediction need (Hair Jr. et al. 2017b). As an example of a research employing PLS-SEM to explore outcome predictors – outcomes that are potentially meaningful to practitioners and may contribute to developing a foundation of entrepreneurship knowledge and theory – consider the study by Criado-Gomis et al. (2018) of how sustainable entrepreneurial orientation within an intrapreneurial context can impact business performance.

Applicable with small samples and non-normal distributions

PLS-SEM is particularly useful when sample sizes are small (i.e., less than 250, Nitzl 2016) or not normally distributed (Hair Jr et al. 2017a). A search of relatively recent entrepreneurship articles produced a plethora of studies with small sample sizes (e.g., Baidoun et al. 2018, $n = 90$; Barba-Sánchez and Atienza-Sahuquillo 2017, $n = 101$; DeTienne and Robb 2016, $n = 102$; Fenech et al. 2019, $n = 23$; Hameed et al. 2018, $n = 72$; Hansen and Nybakk 2018, $n = 89$; Hu et al. 2018, $n = 99$; Ikebuaku and Dinbabo 2018, $n = 70$; Jennings et al. 2016, $n = 163$; Kapinga et al. 2018, $n = 52$; Mohammed 2019, $n = 53$; Sajilan et al. 2019, $n = 170$). Interestingly, none of these studies measured and reported the normality of their sample data, which is consistent with the proposition that researchers often do not assess normality, but rather assume their data is normally distributed (Micceri 1989; Nitzl 2016). Yet, most data applied in business and related social sciences is not normally distributed (Hair Jr et al. 2014; Hair Jr et al. 2019d). Given that entrepreneurship research typically employs small samples, and normality is often not assessed but assumed, PLS-SEM is an excellent fit for this field of study.

Ability to assess complex models

We previously mentioned the possibility of applying “what, when, why, and how” questions, a holistic view, to pave a path toward entrepreneurship theory development. However, entrepreneurship cause-effect relationships are often complex, and PLS-SEM gives researchers the ability to manage that complexity (Sarstedt et al. 2014b). The complexity of discerning cause-effect relationships in entrepreneurship is further exacerbated by dynamic environmental factors such as globalism, changing consumer desires, a continuing wave of new products and services, evolving laws and regulations, cyclical economic trends, and sociocultural changes. In dynamic environments, like entrepreneurship, researchers may apply a “funnel approach” (Richter et al. 2016b, p.379): begin with a plethora of related research questions, take a holistic look, and then narrow the scope through a methodological examination of relationships. Certainly, a funnel approach in entrepreneurship research would generate complex models such as those with five or more constructs, numerous structural path relationships, or constructs with more than six indicators. All of these are consistent with the application of PLS-SEM (Sarstedt et al. 2014b).

In PLS-SEM models, latent constructs are measured with a set of indicators (Hair Jr et al. 2016). For example, researchers might apply a set of indicators to measure each of the five dimensions of entrepreneurial orientation (EO): autonomy, innovativeness, risk taking, proactiveness, and competitive aggressiveness (Lumpkin and Dess 1996). Multiple studies have applied PLS-SEM to measure various EO latent constructs (e.g., Al-Dhaafri et al. 2016; Bouncken et al. 2016; Farsi et al. 2013; Santos et al. 2018; Vega-Vázquez et al. 2016; Williams Jr. et al. 2018). Unquestionably, EO is the more common of the many potential latent constructs commonly examined in entrepreneurship studies. For an example of such a complex model, see Williams Jr et al.’s (2018) model in Fig. 3. This study includes EO as a higher order construct consisting of its five dimensions measured by eighteen indicators, their newly-developed comprehensive strategic approach which is a higher order construct consisting of strategic planning, goal setting, and ratio analysis. These constructs are measured by three indicators each, while firm performance is assessed through subjective satisfaction-based measures. Finally, Williams Jr et al. (2018) consider the interactive effects of EO and CSA and control for firm size in sales and total number of employees.

Capacity to apply formative composite measures

Formative indicators are assumed to cause or form a construct. The opposite is assumed for reflectively measured indicators, which are caused by the construct (Hair Jr et al. 2016). Formative measures are especially relevant to composite constructs representing firm performance, a common dependent variable in entrepreneurship research (e.g., Arzubiaga et al. 2018; Lomberg et al. 2017; Ramadani et al. 2017; Williams et al. 2017; Williams Jr et al. 2018). Moreover, financial numbers and ratios are formative measures of firm performance (Nitzl 2016; Richter et al. 2016a; Rodgers and Guiral 2011). Further, formative measures are appropriate for variables other than financial measures. For example, consider Dijkhuizen et al.’s (2016) study of entrepreneurs’ feelings of success in which the variables “job demands” and “job resources” were measured with formative items.

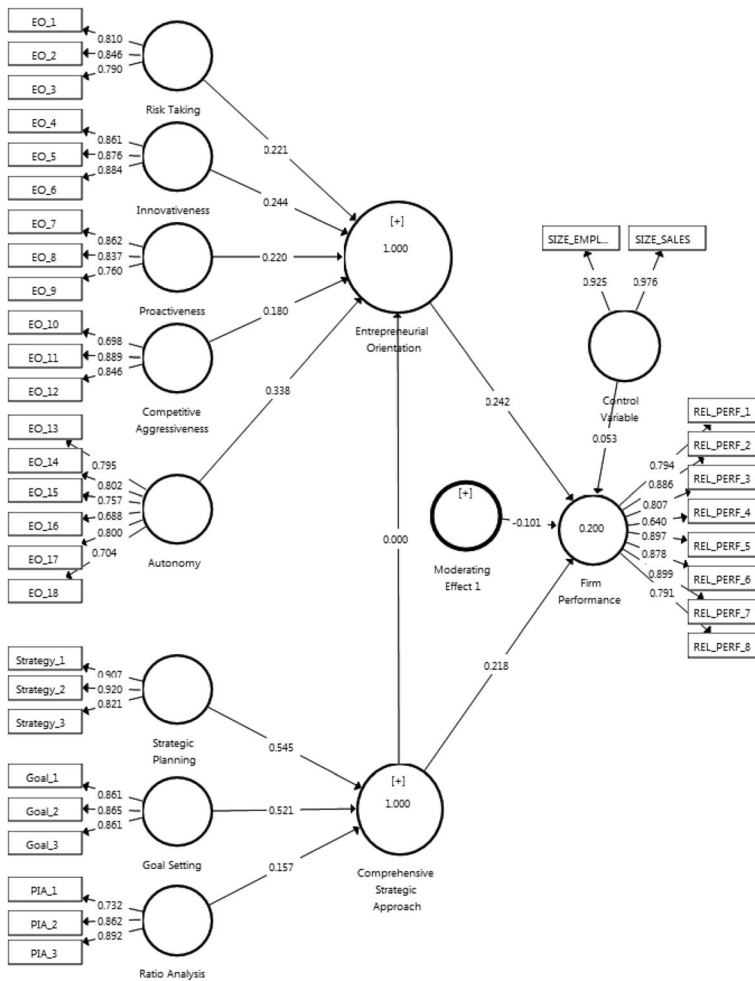


Fig. 3 An example of a complex model applied in entrepreneurship research by Williams Jr. et al. (2018)

In the social sciences, multi-item constructs are frequently used as proxies to measure complex concepts. As noted above, two popular approaches to develop these proxies for concepts or constructs are reflective and formative measurement models. The average summed score method used to include composites in multiple regression introduces bias because it does not remove error, and formative composites create identification problems in CB-SEM (Hair Jr and Sarstedt 2019). Thus, to reduce bias and error in constructs included in theoretical models, researchers should use formative composite measures and PLS-SEM. Applications of PLS-SEM with composite measures have many benefits, including (1) the ability to estimate complex models comprised of many latent and observed variables, and (2) as noted above, to obtain solutions with small sample sizes without identification and convergence concerns (Hair Jr et al. 2019b). These characteristics of PLS-SEM are particularly meaningful since recent studies indicate researchers need to include more observed variables to measure proxies of conceptual variables, regardless of whether they are using common

factor or composite proxies. More indicators reduce factor indeterminacy in common factors and increase the reliability of composite measures (Sarstedt et al. 2016).

Better interpretation for mediation and conditional process models

In recent years, mediation and conditional process assessments exploring analyses such as moderated mediation are increasingly being applied to examine theoretical model effects in an entrepreneurship context, including factors that may influence these effects. When assessing mediation results, some researchers report the results of both structural equation modeling (SEM) and multiple regression using the PROCESS macro (Hayes and Rockwood 2020). However, reporting and comparing the results of both approaches is not necessary since methodological research has demonstrated the limitations of estimating multiple regression models, including mediation, with latent variables (Sarstedt et al. 2020).

A mediation research design proposes a third variable that may facilitate the relationship between an independent and a dependent variable, as shown in Fig. 4. Specifically, a mediation design examines whether a change in the independent variable (X) results in a change of the mediator variable (MED), which in turn changes the dependent variable (Y). Figure 4 displays two examples of such research designs. In both examples the mediation model displays the mediator variable (MED) as theoretically mediating the relationship between X and Y. Both examples also display moderation, or more specifically moderated mediation, a research design in which a third variable (MOD) directly impacts the strength or even the direction of a relationship between two other variables. Thus, the theoretical models illustrate a combination of mediation and moderation in which a moderator impacts one or more direct effects that constitute a mediating effect, and are referred to as conditional process models. The theoretical model on the left is an example of only a single mediated relationship being moderated, while the model on the right displays both mediated relationships being moderated. In short, the models illustrate how theoretical models can be extended to accommodate both mediators and/or moderators.

Many entrepreneurship scholars are likely not aware of the application of PLS-SEM for mediation as well as moderation analyses. Indeed, the superiority of PLS-SEM over other analysis methods was only recently demonstrated (Sarstedt et al. 2020). Among the benefits of using PLS-SEM over PROCESS is that multi-item constructs as components of theoretical models can be assessed easily for both reliability and

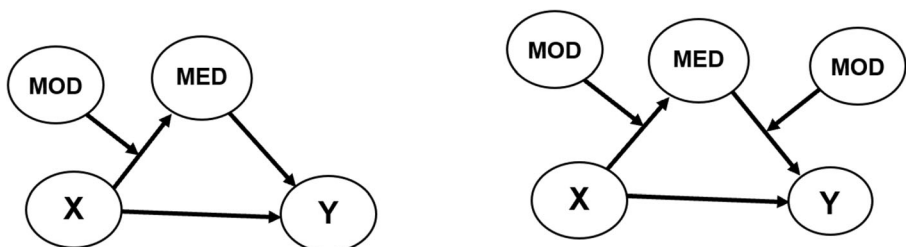


Fig. 4 Examples of moderated mediation

validity, as can both categorical and continuous moderators. Moreover, PLS-SEM more accurately assesses mediated moderation than do multiple regression or covariance-based structural equation modeling approaches. In sum, the composite-based PLS-SEM method is the preferred and superior approach when estimating and reporting the results of mediation and conditional process models, and the PROCESS approach is not needed to assess mediation.

More accurate prediction assessment using holdout samples

Entrepreneurship researchers using multiple regression and PLS-SEM often interpret the R^2 as an accurate measure of their model's predictive power. The R^2 statistic is a useful metric to measure the in-sample explanatory power of regression and PLS-SEM models, but it is not an indication of the model's out-of-sample predictive power (Shmueli 2010; Shmueli and Koppius 2011; Dolce et al. 2017). In-sample prediction is a measure of the ability of a single sample to predict the dependent variable values for the observations in the same sample. Because it uses the same observations to both develop the parameters and to predict the outcome variable observations, the model is optimized to predict the observations of the same sample. Therefore, many researchers believe that R^2 is also a good measure of the ability to predict similar observations that were not included in the original sample. However R^2 is not an accurate measure of out-of-sample prediction and, thus not an accurate approach to infer to the general population.

Recognizing the differences between in-sample explanatory power and out-of-sample predictive power is very important in interpreting quantitative empirical research in general, and entrepreneurship research in particular. Relying only on the R^2 metric to assess a theoretical model's prediction adequacy is problematic. Doing so substantially limits the possibility of generalizing the predictive results of the model to other samples (Pitt and Myung 2002; Sharma et al. 2018; Sharma et al. 2019). The ability to have confidence in the predictive accuracy of theoretical models based on empirical research is a key element in social sciences research, particularly important in the nascent field of entrepreneurship.

In addition to examining the in-sample prediction, entrepreneurship researchers should also estimate the out-of-sample predictive power of their models (Chin 1995; Nitzl and Chin 2017). Estimates of out-of-sample predictive power involve first using an analysis sample (also called a training sample) to estimate the model parameters (Hair Jr. et al. 2019). The analysis sample does not include the observation to be predicted. The second set of observations is called the holdout sample. The parameters of the model developed with the analysis sample are then used to predict the values of the observations in a holdout sample (i.e., the remaining part of the dataset not used for model estimation).

For many years, researchers have proposed the holdout sample approach to assess out-of-sample predictive power as an option with regression, but it is almost never been reported since no software was available to execute the process. To facilitate this type of analysis for PLS-SEM, Shmueli et al. (2016) developed a process called PLSpredict. PLSpredict is a procedure that uses a holdout sample approach to generate observation-level predictions at both an item and a construct level. While standard structural model evaluation metrics such as R^2 and Q^2 provide in-sample predictions, PLSpredict

provides metrics to assess a model's out-of-sample predictive power. We urge entrepreneurship researchers to apply PLS-SEM to examine their theoretical models and to routinely report the results for the PLSpredict procedure.

To assess the predictive power of the theoretical model when using PLSpredict, researchers should apply prediction statistics that measure the prediction error. One prediction metric is the mean absolute error (MAE). This metric measures the average magnitude of the errors in prediction, but does not consider their direction – whether the prediction is over or under. The MAE averages the absolute differences between the predictions and the actual observations using a process in which all the individual differences have equal weight. A second prediction metric is the root mean squared error (RMSE). This metric measures the square root of the average of the squared differences between the model's predictions and the actual measures of the observations. The RMSE metric squares the errors before averaging. Thus, the RMSE statistic assigns a larger weight to larger errors. The approach of the RMSE metric is, therefore, very useful when the analyst wants to avoid large errors, which is often characteristic of business research applications, such as entrepreneurship studies.

To assess PLSpredict error results, researchers should examine the prediction error of the key endogenous construct of their theoretical model. To do so, the key target construct is identified and the Q^2_{predict} statistic is evaluated first. The assessment involves determining whether the model's predictions outperform the most naïve prediction benchmark, which is the means of the analysis sample indicators (Shmueli et al. 2019). The next step is to examine the distribution of the RMSE error statistic. When the RMSE error prediction distribution is highly non-symmetric, the MAE prediction statistic should be used (Shmueli et al. 2019). The process involves comparing the RMSE (or MAE) values with a naïve benchmark. The recommended naïve benchmark is the errors produced by a linear regression model (LM) to generate predictions (Danks and Ray 2018). The process involves comparing the RMSE (or MAE) errors to the LM errors and applying these guidelines (Shmueli et al. 2019):

- If the PLS-SEM prediction errors for RMSE (or MAE) for all indicators are higher compared to the naïve LM benchmark, the model does not have predictive power.
- If a *majority* of the prediction errors (RMSE or MAE) of the dependent construct indicators for PLS-SEM are higher compared to the naïve LM benchmark, the model has low predictive power.
- If a *minority* (or the same number) of the prediction errors of the indicators for PLS-SEM are higher compared to the naïve LM benchmark, the model has medium predictive power.
- If *none* of the prediction errors of the indicators for RMSE (or MAE) for PLS-SEM are higher compared to the naïve LM benchmark, the model has high predictive power.

Advanced approaches from the PLS-SEM tool box

The two advanced PLS-SEM approaches applicable to entrepreneurship research include multigroup analysis, which was recently developed, and hierarchical

component models. In applying multigroup analysis, researchers are able to examine differences in identical models between different specified groups within a data set (Hair Jr et al. 2014; Matthews, Hair Jr and Matthews 2018). For instance, an entrepreneurship researcher may assume all the firms represented in a sample are homogeneous, or the researcher may want to explore how results differ for types of businesses – for instance tech related businesses (apps and IT) versus traditional business or product versus service businesses. As stated above, entrepreneurs operate in a dynamic environment with a multitude of different opportunities. Multigroup analysis, as applied in PLS-SEM, may aid researchers in understanding entrepreneurship in various environments and situations. As an example of multigroup analysis in entrepreneurship research, consider García-Rodríguez et al.'s (2015) study of how different economic and cultural contexts affect entrepreneurial intentions; they applied multigroup analysis to compare Senegal and Spain.

Hierarchical component models (HCM) test measurement structures with two layers of constructs, a higher order component (HOC) and lower order component (LOC), in which LOCs form a HOC (Hair Jr et al. 2019d; Hair Jr et al. 2013; Matthews et al. 2018; Sarstedt et al. 2019). For an example, see Fig. 3 which was explained in the preceding section. Such HCMs may assist entrepreneurship researchers' research and develop knowledge about complex relationships in their field.

Concluding observations

While PLS-SEM is increasingly being applied to estimate structural equation models (Hair Jr et al. 2014; Hair Jr. et al. 2019), it is much less frequently applied in entrepreneurship research. Entrepreneurship researchers need to be familiar with the latest quantitative methodological tools, including when, why and how to apply them. For researchers who have not previously used PLS-SEM, this article provides an overview of relevant and emerging concepts when preparing and finalizing their manuscripts applying PLS-SEM. In addition, for researchers who have applied PLS-SEM in the past, we provide the latest developments and best practices in building PLS-SEM models and interpreting and reporting results. We provide an overview of several recently proposed improvements (PLSpredict and higher order constructs), as well as complementary methods for assessing mediation and conditional process models which we recommend when using PLS-SEM. We also summarize the emergent confirmatory composite analysis (CCA) as an alternative to the use of confirmatory factor analysis (CFA) in the development, adaptation and confirmation of measurement scales. This knowledge can also be useful for journal reviewers and editors to ensure the rigor of manuscripts applying PLS-SEM. Finally, while some methodological researchers have published negative comments about the application of PLS-SEM, several prominent researchers have recently acknowledged the value of PLS-SEM as an analytical technique (Petter 2018). We believe entrepreneurship scholars will be remiss if they do not apply the most recent statistical methods available to explore and better understand the complex phenomena they are researching.

Appendix

Table 4 Entrepreneurship journals found in our search for each keyword with number of studies published applying PLS-SEM for each journal

Key word	Entrepreneurship journals publishing studies utilizing PLS-SEM	Total per Journal
Venturing	<i>Asia-Pacific Journal of Business Venturing and Entrepreneurship</i>	1
	<i>Journal of Business Venturing</i>	4
	<i>Society of Business Venturing</i>	1
Entrepreneur	<i>Inside the Mind of the Entrepreneur</i>	1
Entrepreneurship	<i>Academy of Entrepreneurship Journal</i>	5
	<i>African Journal of Entrepreneurship and Business Management</i>	2
	<i>American Journal of Entrepreneurship</i>	1
	<i>Asia Pacific Journal of Innovation and Entrepreneurship</i>	4
	<i>Covenant Journal of Entrepreneurship</i>	3
	<i>Ekonomski vjesnik/Econviews-Review of Contemporary Business, Entrepreneurship and Economic Issues</i>	2
	<i>Entrepreneurship Education and Pedagogy</i>	2
	<i>Entrepreneurship & Regional Development</i>	2
	<i>Entrepreneurship in International Marketing</i>	1
	<i>Entrepreneurship Theory and Practice</i>	1
	<i>Entrepreneurship and Family Enterprise Research</i>	1
	<i>Entrepreneurship Research Journal</i>	4
	<i>Entrepreneurship and Sustainability Issues</i>	2
	<i>Entrepreneurship-Trends and Challenges</i>	1
	<i>Frontiers of Entrepreneurship Research</i>	2
	<i>Hasanuddin Journal of Applied Business and Entrepreneurship</i>	1
	<i>Indonesian Journal of Business and Entrepreneurship</i>	3
	<i>Innovation Management, Entrepreneurship and Sustainability</i>	1
	<i>International Entrepreneurship and Management Journal</i>	26
	<i>International Journal of Entrepreneurship</i>	9
	<i>The International Journal of Entrepreneurship and Innovation</i>	1
	<i>International Journal of Entrepreneurship and Small Business</i>	2
	<i>International Journal of Gender and Entrepreneurship,</i>	3
	<i>International Journal of Social Entrepreneurship and Innovation</i>	4
	<i>Journal for Business and Entrepreneurship</i>	1
	<i>IUP Journal of Entrepreneurship Development</i>	1
	<i>Journal of Applied Management and Entrepreneurship</i>	1
	<i>Journal of Developmental Entrepreneurship</i>	1
	<i>The Journal of Entrepreneurship</i>	2
	<i>Journal of Entrepreneurship and Business Innovation</i>	1
	<i>Journal of Entrepreneurship in Emerging Economies</i>	8
	<i>Journal of Entrepreneurship and Innovation Management</i>	1

Table 4 (continued)

Key word	Entrepreneurship journals publishing studies utilizing PLS-SEM	Total per Journal
	<i>Journal of Entrepreneurship, Management and Innovation</i>	2
	<i>Journal of Entrepreneurship & Organization Management</i>	1
	<i>Journal of Entrepreneurship and Business</i>	6
	<i>Journal of Entrepreneurship Education</i>	4
	<i>Journal of Global Business and Social Entrepreneurship</i>	1
	<i>Journal of Global Entrepreneurship Research</i>	8
	<i>Journal of Innovation and Entrepreneurship</i>	2
	<i>Journal of International Business and Entrepreneurship Development</i>	3
	<i>Journal of International Entrepreneurship</i>	3
	<i>Journal of Research in Marketing and Entrepreneurship</i>	2
	<i>Journal of Entrepreneurship, Business and Economics</i>	1
	<i>Journal of Small Business & Entrepreneurship</i>	4
	<i>Journal of Small Business and Entrepreneurship Development</i>	1
	<i>International Journal of Gender and Entrepreneurship</i>	1
	<i>Strategic Entrepreneurship Journal</i>	1
	<i>Sustainability in Innovation and Entrepreneurship</i>	2
	<i>Technology Transfer and Entrepreneurship</i>	1
	<i>The Role and Impact of Entrepreneurship Education</i>	2
	<i>The Southern African Journal of Entrepreneurship and Small Business Management</i>	1
	<i>World Journal of Entrepreneurship, Management and Sustainable Development</i>	10
Entrepreneurial	<i>Entrepreneurial Business and Economics Review</i>	1
	<i>International Journal of Entrepreneurial Behavior & Research</i>	6
	<i>International Journal of Entrepreneurial Knowledge</i>	2
	<i>International Journal of Entrepreneurial Research</i>	2
	<i>Total</i>	173

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